THE CATHOLIC UNIVERSITY OF AMERICA

SCHOOL OF ENGINEERING

ENGINEERING MANAGEMENT PROGRAM (EMP)

CMGT 570: PROJECT MANAGEMENT

The SIM Project® final team report

Team members:

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Introduction:

This report evaluates the performance of the team during the Alliance Prototype Project by comparing actual outcomes against the established goals, baseline budget, and schedule. It outlines the team's composition, initial strategy, actual versus planned performance, and key lessons learned throughout the project. The report also offers recommendations for alternative strategies and improvements for future simulations. The team, comprised of Keerthana Gandla, Akhil Munigonda, Vijay Madda, Akshitha Pitla, and Mrudhula Pydipalli, completed three project runs, each exploring different resource selection and operational strategies.

Team Identification:

- **Keerthana Gandla** is pursuing a master's degree in Data Analytics and has previous work experience in the field of data analysis.
- **Akhil Munigonda** is pursuing a master's degree in Data Analytics and brings prior experience in business intelligence and analytics.
- **Vijay Madda** is pursuing a master's degree in Computer Science with a background in technology consulting sector.
- **Akshitha Pitla** is pursuing a master's degree in Data Analytics with a background in software development.
- **Mrudhula Pydipalli** is pursuing a master's degree in Data Analytics with a background in computer science and application development.

Each team member contributed unique skills and expertise, enabling the team to effectively execute and manage the project.

Original Strategy:

The team developed an original strategy for each of the three runs. For the first two runs,

Keerthana Gandla explains her original strategy as follows:

The original strategy for the first run was to complete the project within the set goals of a $50,000 budget, 11 weeks duration, and no more than 12 defects I focused on becoming familiar with the simulation software and initially followed standard resource allocation guidelines. However, the project faced setbacks due to the absence of David and Chandra Lake, which impacted resource availability and delayed critical tasks.The project faced several early challenges, such as scheduling conflicts, poor resource allocation, and delays caused by task

dependencies—particularly the Network Diagram, which was repeatedly postponed and ended up holding back progress for several team members.

Some inefficiencies occurred because workers were assigned to tasks before their dependencies were fully completed. This led to higher costs, reduced productivity, and quality challenges. As a result, the project extended to 12.6 weeks, exceeding the original 11-week goal. Although 12 defects were recorded meeting the defect target Task 6: Customer Acceptance Test remained incomplete due to quality issues, it was done only 96%, worsened by Chandra Lake's absence during a critical phase.

Financially, the final project budget reached $53,933, which was $3,933 over the target. However, this overrun was within 10% of the original budget, which is generally considered an acceptable range in project management. While non-labor costs stayed under budget, labor cost overruns and the impact of absent resources highlighted the need for better task sequencing, contingency planning, and tighter control over project execution.

For the second run, the strategy was to optimize the approach from the first run and improve final results, aiming for a variance within 10% of the original goals. I decided to experiment with new strategies, including assigning only one software engineer instead of two, to test resource efficiency. This approach, while innovative, introduced additional risks to project execution and quality control.

Throughout the project, there were notable challenges. David Lewis was released early from the project, reducing available IT support. Task 6 remained incomplete, due to quality issues directly impacting the project's final acceptance.

Additionally, Task 4 experienced delays because no workers were assigned, and thus no work could be completed on that task during a critical period. To help address schedule pressures, David Lewis, Jane Hodges, and Sally Nicoles voluntarily worked overtime. However, Jane Hodges was later absent, further straining resource availability. By week 14, there was no new staffing activity, limiting the ability to accelerate remaining tasks.

The project ultimately finished in 14.4 weeks, exceeding the original 11-week goal. Defects also increased significantly: although the original target was 12 defects, the project accumulated nearly 14 defects .Financially, the final actual budget was $57,450, exceeding the $50,000 target by $7,450. This represents a 15% overrun, The overrun was driven mainly by increased labor costs ($55,950 actual vs. $40,160 planned), although non-labor costs remained well under control.

Key indicators like the Projected Finish (week 14.4) and PPI (Project Performance Index), which dipped below 1.0 early on before stabilizing later, reflect the impact of early resource allocation issues and worker shortages.

Lessons from this run will be critical for refining resource strategies and improving both quality and schedule adherence in future project simulations.

Akhil Munigonda explains his original strategy as follows:

The first run of the Sim Project was approached as a learning opportunity, with the goal of completing the simulation within the given constraints while exploring the software and project dynamics. During this initial attempt, I familiarized myself with the simulation tools, task structure, and resource management processes. I followed the provided task descriptions to assign workers and made budgeting decisions according to the given guidelines. However, managing the simulation week by week quickly highlighted the complexities of project execution.

The total project cost reached $91,114, with a recorded defect count of 13 than the expected 12. The project spanned 21.4 weeks and while most tasks were completed in full, Task #6 remained slightly incomplete at 93.6%. Despite this, the final project performance was marked as 100% complete. However, it ended with a significant budget overrun . One key mistake during the run was assigning workers to tasks before their dependent tasks were completed. Moreover , the unavailability of David reduced resource capacity, hindering progress on key tasks and affecting overall timelines. Ongoing delays with the Network Diagram task created a bottleneck that disrupted the workflow and held up progress for several team members.

Quality tracking further highlighted persistent issues. Defect counts increased steadily across software and hardware components, with total project defects peaking in Weeks 17–19. Software defects in particular surpassed the target quality threshold, reflecting gaps in quality assurance practices.

This led to reduced productivity and increased costs. As a result, more resources from the reserve budget had to be used, and this contributed to quality issues later in the project. These decisions ultimately increased costs and lowered efficiency.

For the second run, the strategy was to optimize the first attempt and improve the final results to stay within a 5% variance.For the second run, the strategy was to optimize the first attempt and improve the final results to stay within a 5% variance. the final actual cost rose to $86,218, reflecting a substantial budget overrun when compared to the planned and earned values of $50,000 each. The project duration spanned 14.4 weeks.

From a quality perspective, defect trends showed improvement at end 13total defects were recorded exceeding the expected count of 12. with fewer software and hardware-related issues compared to the previous run. Some issues persisted, such as task dependencies preventing Chandra Lake, Jane Hodges, and Tea Dominque from making progress, which led to labor costs being charged without work completion. Additionally, David Lewis was absent, which affected

the schedule slightly. Despite these challenges, resource allocation was more efficient this time. For example, although Lisa Derring was requested but initially unavailable, we chose to wait rather than replace her, which ultimately worked out since she remained an available resource.

To conclude ,the lessons learned during this run—especially regarding task dependencies, budgeting discipline, and the impact of early decisions—proved invaluable in shaping more efficient strategies in subsequent runs. Recognizing the consequences of misaligned scheduling and improper resource allocation laid the groundwork for improvement and more effective project control in final attempt.

Vijay Madda explains his original strategy as follows:

The goal for the first run was to complete the project within a $50,000 budget, finish within 11 weeks, and maintain controlled defect levels. In the first run of the project, I encountered several major challenges that led to eventual failure. Initially, execution was steady, but staffing shortages, scheduling conflicts, and poor dependency management — especially with delays in completing the Network Diagram — began to cause serious setbacks. Rising defect rates forced the urgent creation of **Task 9: Emergency Defect Correction** to address quality issues. However, from Week 13 onward, progress stalled completely because Task 9 was blocked by unfinished predecessor work. Despite management interventions and warnings, no further progress was made for 13 consecutive weeks. Ultimately, the project was officially shut down at 88.89% completion after Week 24, far exceeding the original schedule and budget. Key problems included lack of contingency planning, resource mismanagement, and unresolved task dependencies.

In the second run, after consulting with the professor and getting valuable guidance from the TA, I approached the project with better planning and task management. While some dependency and staffing issues still occurred, early identification of risks and stronger communication helped move the project forward. Defects, which had peaked around Week 8, were managed more proactively, and even though reserve overspending still occurred, all tasks were eventually completed. Strategic staff releases, voluntary overtime, and more efficient task assignments contributed to success. The project finished at 100% completion, although the final cost reached $66,428, exceeding the initial $50,000 budget by about 32%.

In comparison, the second run showed clear improvements in execution, team communication, defect management, and recovery from delays. Unlike the first run, where the project failed to finish, the second run demonstrated stronger leadership, better risk mitigation, and successful delivery, even with some remaining budget challenges. The lessons learned between the two runs highlighted the importance of proactive dependency management, fast decision-making, and strong stakeholder engagement for future projects.

In conclusion, the second run of the project was significantly more successful, showing how applying early lessons, improving communication, and focusing on critical task flow can lead to better outcomes, even in challenging conditions.

Akshitha Pitla's experience with the project is as follows:

The goal for the first run was to complete the project while staying within the planned budget, meeting the expected timeline, and maintaining an acceptable level of quality by minimizing defects.In the first run of Sim Project, I encountered several significant challenges that hindered progress. Initially, staffing shortages and resource mismanagement were major issues. With David Lewis as the only available resource, progress on Task 1 was slow, and by Week 1, it was only 55.32% complete. The situation worsened in Week 2 when a key stakeholder from the Marketing team missed an important design approval meeting, resulting in missed requirements that expanded the project scope unexpectedly. This scope creep added additional tasks, further straining an already tight schedule. Throughout the early weeks, absenteeism, including David Lewis's, created further delays and resource inefficiencies. Task dependencies weren't managed well, leading to workers being assigned to tasks that couldn't proceed, which drained 75% of the risk reserves by Week 4. By Week 5, morale issues began to surface, and rising defect rates, coupled with stakeholder dissatisfaction, especially from Marketing, placed considerable pressure on the project. Despite these hurdles, the team managed to complete the project, delivering it at 100% completion. However, the final cost of $60,554 exceeded the original budget by 21%, primarily due to unforeseen scope changes, resource mismanagement, absenteeism, and defect management issues. While the core functional requirements were met, the project was marked by a substantial budget overrun, a delayed timeline, and persistent quality control challenges.

In the second run, I was better equipped to manage the issues that had affected the first run. Although similar staffing conditions existed, the lessons learned from the first run allowed me to manage tasks more efficiently from the start. Despite initial absenteeism issues, notably with David Lewis, and a delay in completing Task 1, the team pushed forward with a more focused approach. A key improvement was stakeholder communication. Although some stakeholders still missed key meetings, the project team proactively kept everyone informed and avoided allowing delays to escalate. Task dependencies were addressed more effectively, ensuring that workers were assigned to tasks that could proceed, and the reserve fund depletion was not as severe as in the first run. Defect management became a central focus in the second run, leading to the introduction of an emergency defect correction task in Week 10. While this task led to some cost overruns, it helped manage quality control issues and prevent further delays. The team also embraced voluntary overtime and a more focused approach to completing tasks. By the end of the project, the final cost was $54,500, reflecting a 9% budget overrun compared to

the original $50,000. This was a much smaller overrun than the first run's 21%, indicating better cost management. The second run achieved 100% completion with fewer disruptions, better communication, and more effective task management. While challenges remained, the team's ability to adapt, make strategic decisions, and prioritize critical tasks allowed us to deliver a successful project despite lingering resource allocation issues and cost overruns.

In comparison to the first run, improvements in the second run were clear. The Project Performance Index (PPI) in the second run showed quicker recovery and more consistent efficiency, while the first run experienced longer periods of inefficiency. The Estimated Project Finish (EPF) was pushed to Week 15-16 in the second run due to defects and rework, compared to the earlier delay in the first run. Cost overruns were more controlled in the second run, with the final budget overrun at $54,500 (9%) compared to the first run's $60,554 (21%). Defect management was also more effective in the second run, with defects peaking later and being corrected more efficiently.

In conclusion, the second run of SimProject showed notable improvements in stakeholder engagement, resource management, and defect resolution. I learned from the early mistakes in the first run and applied more proactive measures to address communication breakdowns and task dependencies. While absenteeism still posed challenges, the team was better able to manage these risks, ultimately delivering the project with a more controlled budget overrun and higher-quality results. The lessons from both runs will be invaluable for optimizing resource strategies and improving both quality and schedule adherence in future project simulations.

Mrudhula Pydipalli explains her original strategy as follows:

In Run-1 of the Sim Project, the project got off to a challenging start. During Week 1, only one task was active and was 54.05% complete by the end of the week. David Lewis, the sole resource working at the time, was highly productive, contributing 40 hours with 108% productivity. However, the project was already lagging budget expectations with an actual cost of $1,720 versus a planned value of $2,609 and earned value of $2,350. Despite being early in the schedule, this discrepancy indicates early inefficiencies. Project completion stood at only 4.7%, setting the tone for a recovery-focused effort in subsequent weeks. Key meetings were scheduled, including invitations for the design approval meeting, and leadership emphasized the critical importance of completing the project on time and within budget to support the future Alliance project. Additional resources were expected to arrive in upcoming weeks to boost project momentum.

Run-2 of the project, encountered several early setbacks, including scheduling issues, resource mismanagement, and dependency delays—most notably the repeated delays related to the Network Diagram task, which stalled progress for multiple team members. Despite these obstacles, the team steadily improved over the 12-week duration. Quality defects peaked around week 8, prompting leadership to pressure the team to reduce open issues below the project's acceptance threshold. By leveraging team commitment and voluntary overtime, tasks progressed effectively, and defect levels began declining, avoiding the need for an additional emergency correction task. The project ultimately achieved full completion in Week 12, with all seven tasks marked 100% complete. The final metrics showed the project slightly over budget ($52,976 actual vs. $50,000 planned) but matched the earned value target. Overall, while the project faced notable leadership and communication challenges early on, it finished successfully due to adaptive management, strategic reassignment, and dedicated teamwork.

In Conclusion Despite a rocky start and several project management missteps, Run-2 concluded successfully. Strong mid-project interventions, enhanced leadership support, and a dedicated team effort helped recover the schedule and control defect levels. Key lessons include the importance of early stakeholder communication, timely task sequencing, proactive resource management, and continuous quality monitoring. With these takeaways, future runs can expect smoother execution and better cost control.

Third Run Strategy and Process:

In the third run, we collaborated as a team, with each member given three opportunities to apply insights gained from the previous two runs to optimize our results. Our objective was to complete the project within 5% of our initial goals. Working as a group provided several benefits, including sharing lessons from earlier runs, learning from past mistakes, exchanging tips to improve performance, and streamlining project completion.

Through group discussions, we identified the most effective strategies used by different project managers and merged them into a unified, optimized plan. By analyzing previous missteps, we refined our approach and focused on improving efficiency. Tools such as the Gantt chart and Network Diagram helped us strategically assign resources based on individual strengths and anticipated delays in early tasks.

We also improved resource allocation by assigning team members to tasks based on relevant experience rather than job titles. This flexibility allowed us to finish tasks ahead of schedule. Unlike in the first run, where idle workers were released, we instead reassigned them more effectively in this run, resulting in better resource utilization and reduced overall costs.

The primary objective for the third run was to complete the project within 5% of the overall goals. The final project cost was $49,802. The project was completed in 62 days which exceeded

the target of 55 days and recorded only 10 defects, remaining well below the maximum allowable limit of 12. which stayed within the acceptable 10% variance. These results reflect strong overall performance in budget, timeline, and quality.

However, in one case, the project duration extended to 12.4 weeks (approximately 62 days), which exceeded the target of 55 days. This delay was caused by a holdup in Task 2, which had a downstream impact on dependent activities. Documentation Review (Task 7) was also slightly delayed. Omar Mosa's assigned task could not begin until a predecessor task was completed, as in the Network Diagram. During this period, no productive work could be done, but labor costs continued to accrue, being charged to Reserves.

Once his task was unblocked, Omar Mosa voluntarily worked overtime to help the team recover from the schedule slippage. His commitment and extra effort were instrumental in reducing the overall delay and ensuring the project remained within acceptable performance boundaries. His contribution highlights the value of engaged and proactive team members in overcoming project challenges.

Project Performance Overview Across Three Attempts

In our **first run**, the goal was to complete the project within 55 days, stay within 15% of the $50,000 budget, and limit quality defects to fewer than 12. However, due to limited experience, poor resource allocation, and breakdowns in communication, we were unable to meet these targets. The project ran over budget and beyond the planned timeline, and the number of defects was higher than acceptable, leading to further delays and increased costs.

For the **second run**, we took a more structured approach by reassessing our strategy. We kept our targets flexible, aiming to finish the project within 15% of both the time (55 days) and budget ($50,000), while still maintaining defect levels under 12. Through improved resource planning, clearer communication, and a more thoughtful execution strategy, we successfully achieved these revised objectives.

The **third run** represented our most refined and collaborative effort. Drawing from lessons learned, we set a tighter performance target: completing the project within 5% of our original goals. As a team, we implemented a combined strategy based on each member's strengths and experiences from previous runs. The final cost was $49,802 only 0.4% under $50,000 estimated budget. The project recorded 10 defects, well under the maximum limit of 12. However, in one case, the project extended to 12.4 weeks (approximately 62 days), exceeding the 55-day timeline.

The delay stemmed from a bottleneck in Task 2, Documentation Review (Task 7) was also slightly delayed which held up Omar Mosa's dependent task. During the wait, no progress was made, yet labor costs were charged to Reserves. Once the task was unblocked, Omar voluntarily worked overtime, helping recover lost time and contributing significantly to the project's overall success.

Conclusion

Our three-run progression demonstrates the importance of continuous improvement, data-driven planning, and teamwork. While the first attempt highlighted our initial weaknesses, it laid the groundwork for the significant improvements made in the second and third runs. The final run, despite a minor timeline overage, successfully met cost and quality goals—showcasing how collaboration, adaptability, and individual commitment can drive project success.

Project Outcome and Performance Graphs:

The Project Status Report shows the project is 100% complete, delivered under budget with an actual cost of $49,802. Both the planned and earned values were $50,000, indicating efficient cost and schedule performance.

Project Status Report

The Project % complete	100%
Actual Cost	$49,802
Planned Value	$50,000
Earned Value	$50,000

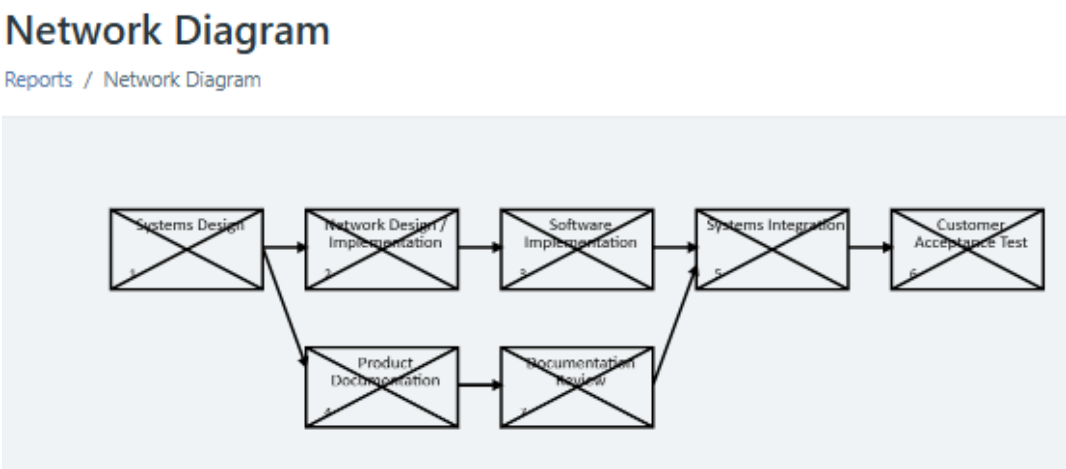
Task Status Report shows that all 7 tasks are 100% complete, indicating the project was fully executed with no pending work remaining.

Task Status Report

Task #1 is: 100% complete
Task #2 is: 100% complete
Task #3 is: 100% complete
Task #4 is: 100% complete
Task #5 is: 100% complete
Task #6 is: 100% complete
Task #7 is: 100% complete

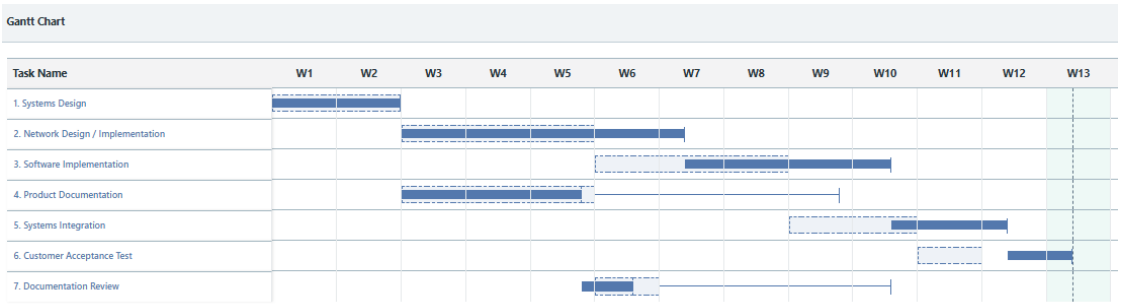
Network Diagram Summary

The network diagram outlines the project's task dependencies. It shows the logical sequence from **Systems Design** to **Customer Acceptance Test**. Critical dependencies are highlighted—particularly that **Systems Integration (Task 5)** cannot begin until both **Software Implementation (Task 3)** and **Documentation Review (Task 7)** are completed. This structure clearly illustrates how delays in early tasks like **Network Design** can cause cascading impacts on the project timeline.

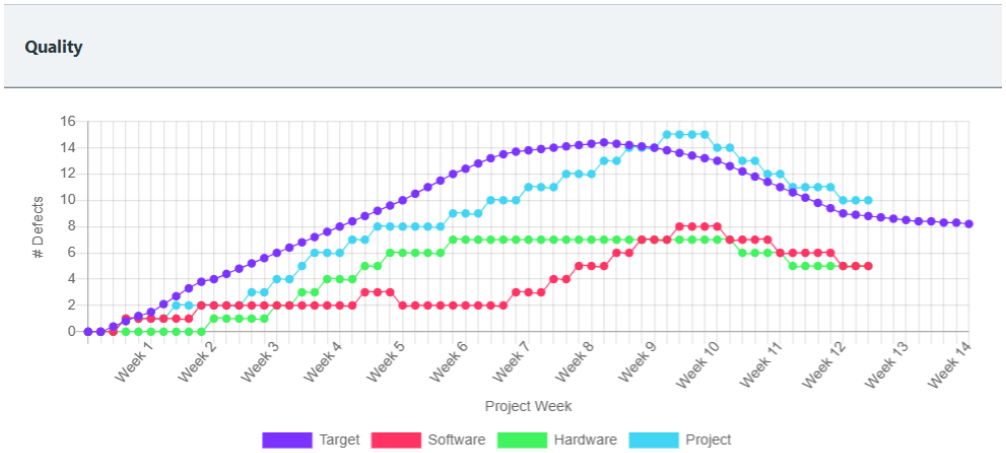


Gantt Chart Summary

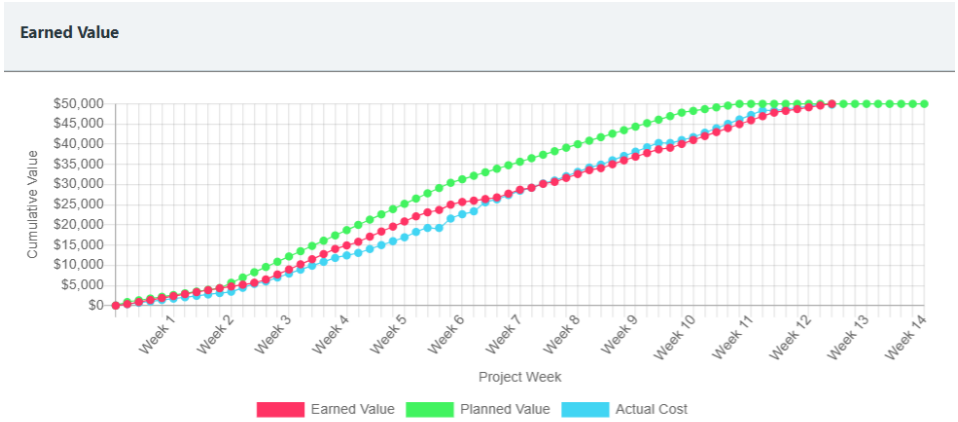
The Gantt chart shows the actual and planned timelines of all project tasks. Notably, delays occurred in **Network Design/Implementation (Task 2)**, which pushed back **Software Implementation (Task 3)** and ultimately extended the project to **Week 12.4**. **Documentation Review (Task 7)** was also slightly delayed. These overlaps and dependencies visually confirm the reason for the extended project duration.



The chart tracks the number of defects across the project timeline by category: **Software (red)**, **Hardware (green)**, and **Overall Project (blue)**, compared to the **target threshold (purple)**. project defects peaked near Week 9 but dropped to **10 by Week 12.4**, staying below the target of 12. Software defects rose mid-project, while hardware defects remained steady, indicating effective overall defect control.



This chart shows the project's **earned value**, **planned value**, and **actual cost** over time. By Week 12, the earned value and actual cost both reached approximately **$49,802**, aligning closely with the **planned value of $50,000**. This indicates that the project was delivered nearly on budget and on schedule, reflecting strong cost and schedule performance.



Lessons Learned

1. **Value of Experience and Proper Resource Allocation**
    - The first run showed that lack of experience and poor resource planning led to missed goals.
    - Assigning tasks based on skills and experience is crucial for efficient project execution.
2. **Flexibility in Goals and Strategies**
    - After underperforming in the first run, the team adjusted targets and strategy in the second run.
    - Being open to change based on past performance helped improve results.
3. **Teamwork and Communication Are Key**
    - In the third run, close collaboration and knowledge sharing led to better performance.
    - Effective communication enabled the team to align on goals and resolve issues faster.
4. **Set Ambitious but Achievable Goals**
    - In the third run, the team aimed to complete the project within 10% of the original targets.
    - This encouraged high performance while keeping expectations realistic.
5. **Smart Task Assignment Enhances Efficiency**
    - Resources were assigned based on expertise, not job titles.
    - This allowed for early task completion and better use of available skills.
6. **Continuous Monitoring and Quality Checks**
    - Mid- and late-stage quality reviews helped detect and resolve defects early.
    - Ongoing monitoring ensured higher quality and reduced rework.
7. **Engage Stakeholders Effectively**
    - Regular updates and involving stakeholders in key phases ensured alignment.
    - Clear communication helped manage expectations and strengthened trust.

Recommendations

1. **Invest in Training and Skill Development**
   Provide team members with continuous learning opportunities to enhance their technical and project management skills. Training and mentoring sessions can improve confidence, reduce errors, and increase overall efficiency—especially for less experienced team members.

2. **Enhance Resource Planning and Allocation**
   Use data-driven tools and assessments to match resources to tasks based on expertise, availability, and workload. Proper allocation not only optimizes productivity but also minimizes delays and unnecessary costs caused by idle or misaligned resources.

3. **Promote Open Communication and Cross-Functional Collaboration**
   Create a culture that encourages transparency, active listening, and feedback. Regular team meetings and open channels of communication help resolve issues early, align goals, and foster a stronger team dynamic.

4. **Adopt an Iterative and Flexible Planning Approach**
   Rather than rigid, static planning, implement rolling-wave or adaptive planning techniques. Set short-term milestones that can be adjusted based on project progress and performance, allowing for real-time corrections and responsiveness to change.

5. **Implement Comprehensive Quality Assurance**
   Establish clear quality benchmarks and conduct regular reviews throughout the project lifecycle. Early detection of defects and ongoing quality monitoring reduce rework, maintain standards, and ensure the final product meets stakeholder expectations.

6. **Strengthen Stakeholder Engagement and Transparency**
   Involve stakeholders from the outset and maintain consistent updates throughout the project. Active participation in design reviews, status meetings, and delivery milestones ensures alignment, trust, and fewer last-minute changes.

7. **Foster Continuous Improvement and Knowledge Sharing**
   Conduct post-project reviews to capture lessons learned, and maintain a knowledge base for future reference. Encourage teams to share insights, best practices, and solutions to recurring challenges to build institutional knowledge.

8. **Encourage Adaptability and Build Resilience**
   Equip teams with the tools and mindset to adapt to unexpected changes or delays. Promote a solution-oriented culture that views setbacks as learning opportunities, helping teams remain focused and proactive under pressure.

Conclusion

The progression across the project runs highlights the critical value of continuous improvement, strategic flexibility, and strong project management. Although the first run encountered significant setbacks, the team applied those lessons to refine their approach in subsequent runs—setting more attainable goals, improving planning, and enhancing execution. Through better resource allocation, stronger collaboration, open communication, and consistent quality control, the team was able to significantly improve performance and exceed expectations in later phases. Looking ahead, organizations should prioritize workforce development, adopt structured planning and quality assurance frameworks, promote transparent communication, support knowledge sharing, and embrace adaptability to ensure sustained project success.